# Advanced Computer Architecture CMP 511

## Multithreaded Architecture

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# Thread

- In computer architecture, a thread is a sequence of instructions that runs within a program. Threads are the basic unit of execution for a CPU.
- The operating system allocates processor time to threads.
- Threads can run in parallel on a multi-core CPU.
- Threads can share resources like memory, executable code, and variables.

# Multithreading

- In computer architecture, multithreading is when a central processing unit (CPU) can run multiple threads simultaneously. Each thread is a separate flow of control that runs a process.
- CPU has the ability to execute multiple threads within an execution, which is supported through operating system
- How does multithreading work?
  - A processor can switch between threads, allowing them to execute concurrently.
  - Threads can run independently but share process resources.
  - Depending on the hardware, threads can run in parallel if they are distributed to their own CPU core.

# Multithreading

- **Benefits of multithreading**
  - Since more than one thread or more than one instructions, are simultaneously executed within the processor, it will definitely improve the performance of the compute. Hereby the output gets generated fast.
  - It can also handle multiple requests from the same user
  - Increased throughput, utilization of multiprocessor architecture,  economical
- **Disadvantages of multithreading**
  - It needs more careful synchronization
  - It imposes context switching overhead
  - Consumes a lot of CPU/memory. In fact multiple threads they do interfere when they use the shared hardware resources. E.g. cache, Translation lookaside buffer (TLB)

# Latency Hiding Techniques

- **Data prefetching techniques**
  - The goal of prefetching is to make available in the cache before the consumer places its request. For example, when the user enters their search query, the search engines goes ahead and delivers the result to the user
  - So based on which results a user typically visits (usually the 1st or 2nd) the resources for these pages are then prefetched using in a faster loading time if the user clicks the link
- **Use of coherent caches**
  - In large scale computers, the cache are distributed and made coherent by implementing cache coherence protocols
- **Use of relaxed memory consistency models**
  - **Weak Consistency:**
    - Does not guarantee that all operations appear in a total order
    - Requires explicit synchronization operations to ensure consistent data access between threads.
  - **Processor Consistency (PC):**
    - Relaxes ordering rules compared to strict sequential consistency by allowing a processor to read from its cache before a previous write operation is fully committed to memory.
    - Still maintains ordering within a single processor.
  - **Release Consistency:**
    - Builds upon weak consistency by distinguishing between "acquire" and "release" synchronization operations.
    - Guarantees that all writes performed before a "release" operation become visible to other threads once they perform an "acquire" operation.
- **Use of multithreading to do context switching**
- **Use of overlapping operation during latency**
  - Operation of remote memory access is overlapped with another operation by the processor, means the processor need not wait till the remote data is accessed

# Multithreading Principles

- Latency (L)
- The number of threads (N)
- The context switching overhead (C)
- The interval between the switches (R)

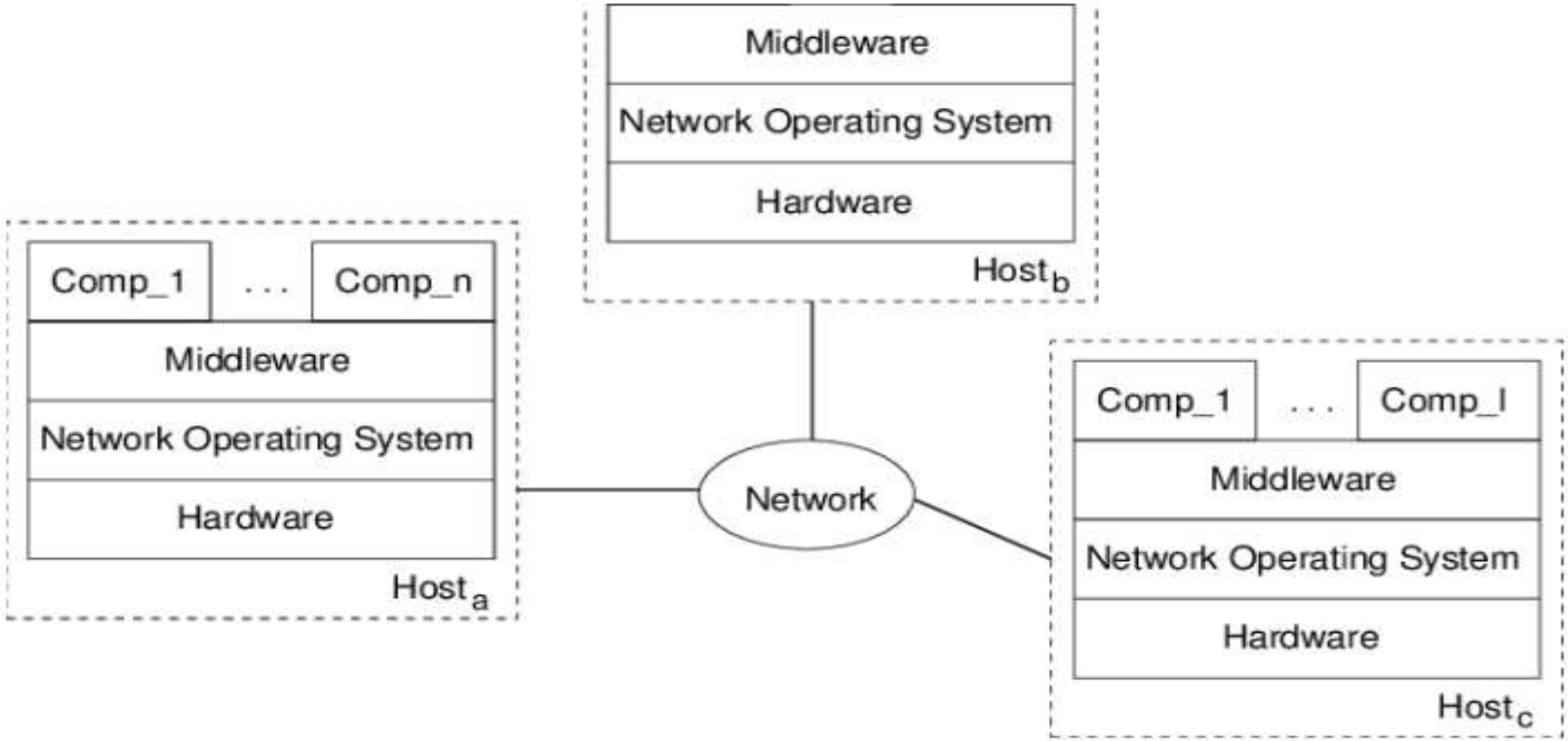
# Multithreading Architectures

- Von-Neumann model
- Data flow model
- Hybrid Model

# Distributed System

- A distributed system is a collection of independent/autonomous computers/host linked by a computer network that  appears to the users of the system as a single coherent system
- Each host execute components and operates a distribution middleware.
- Middleware enables the components to coordinate their activities
- User perceive the system as a single, integrated computing facility

# Distributed System



# Centralized versus Distributed System

| Centralized System | Distributed System |
| --- | --- |
| 1) Centralized system have non-autonomous components | 1) Distributed system have autonomous components |
| 2) Distributed system are build using homogenous technology | 2) Distributed system are build using heterogenous technology |
| 3) Multiple users share the resources of a centralized system at all times | 3) Distributed system are executed in concurrent process |
| 4) Centralized system have a single point of control and of a failure | 4) Distributed system have multiple point of failure |

# Examples of Distributed System

- Internet is the largest distributed system in the world. World Wide Web is the largest application running on the internet. It becomes the most popular distributed software application ever created

a) Video on demand
   - ✓Client components are used to display videos
   - ✓Clients are hosted in the homes of the customers
   - ✓Server components load and transmit videos
   - ✓Multiple servers are needed in order to scale

b) LAN and Intranet

c) Database management system and so on

# Types of Distributed System

a) Distributed Computing System
   - ✓ **Cluster computing**, **Grid Computing** etc.

b) Distributed Embedded System
   - ✓ Home system, health care system, sensor network etc.

c) Distributed Application System
   - ✓ Transmission Processing System (TPS), Enterprise Application System (EAS) etc.

# Advantages of Distributed System

- **Reliability:** if one machine crashes, the rest of the computers remain unaffected and the system can survive as whole
- **Shareability:** sharing of resource: shared data is required to many applications such as banking, reservation system and computer supported cooperative work
- **Scalability:** new machines can be added incrementally as per the requirements as processing power grow.
- **Open-System:** this is the most important characteristic in the distributed system. Since it is an open system, it is always ready to communicate with other system. An open system that scales has an advantage over a perfectly closed and self-contained system
- **Speed:** a distributed system may have more total computing power than a mainframe

**Others**

- Local autonomy
- Potential cost reduction

# Disadvantages of Distributed System

- **Security:** the easy distributed access in distributed computing system increases the risk of security. The sharing of data creates problem of data security
- **Troubleshooting:** troubleshooting and diagnosing problems are the most important disadvantages of distributed system. The analysis may require connection to remote nodes or checking communication between nodes
- **Software:** because of more software components that comprise a system, there is a chance of error occurring
- **Networking:** the underlying network in distributed computing system can cause several problems such as transmission problem, overloading, loss of messages. Hence, the problems created by network infrastructure are the disadvantages of distributed computing.

**Others**

- Complexities
- Multiple point of failure

# Cluster Computing

- When two or more computers are used together to solve a problem, it is considered as computer cluster.
- It consists of many of the similar type of machines.
- They are tightly coupled using dedicated network connections.
- All machines share resources such as common home directory
- It must have software as an MPI implementation installed to allow programs to run across all the nodes.

# Types of Clusters

- High Performance Clusters (HPC)
- High Throughput Clusters (HTC)
- High Availability Clusters (HA)
- Load Balancing Clusters (LBC)
- Hybrid Clusters

  Eg. HPC+HA

# Grid Computing

- Grid computing is a computing infrastructure that combines computer resources spread over different geographical locations to achieve a common goal.
- All unused resources on multiple computers are pooled together and made available for a single task. Organizations use grid computing to perform large tasks or solve complex problems that are difficult to do on a single computer.
- For example, meteorologists use grid computing for weather modeling. Weather modeling is a computation-intensive problem that requires complex data management and analysis.
- Processing massive amounts of weather data on a single computer is slow and time consuming. That’s why meteorologists run the analysis over geographically dispersed grid computing infrastructure and combine the results.

# Why is grid computing important?

Organizations use grid computing for several reasons.

- **Efficiency**: With grid computing, you can break down an enormous, complex task into multiple subtasks. Multiple computers can work on the subtasks concurrently, making grid computing an efficient computational solution.

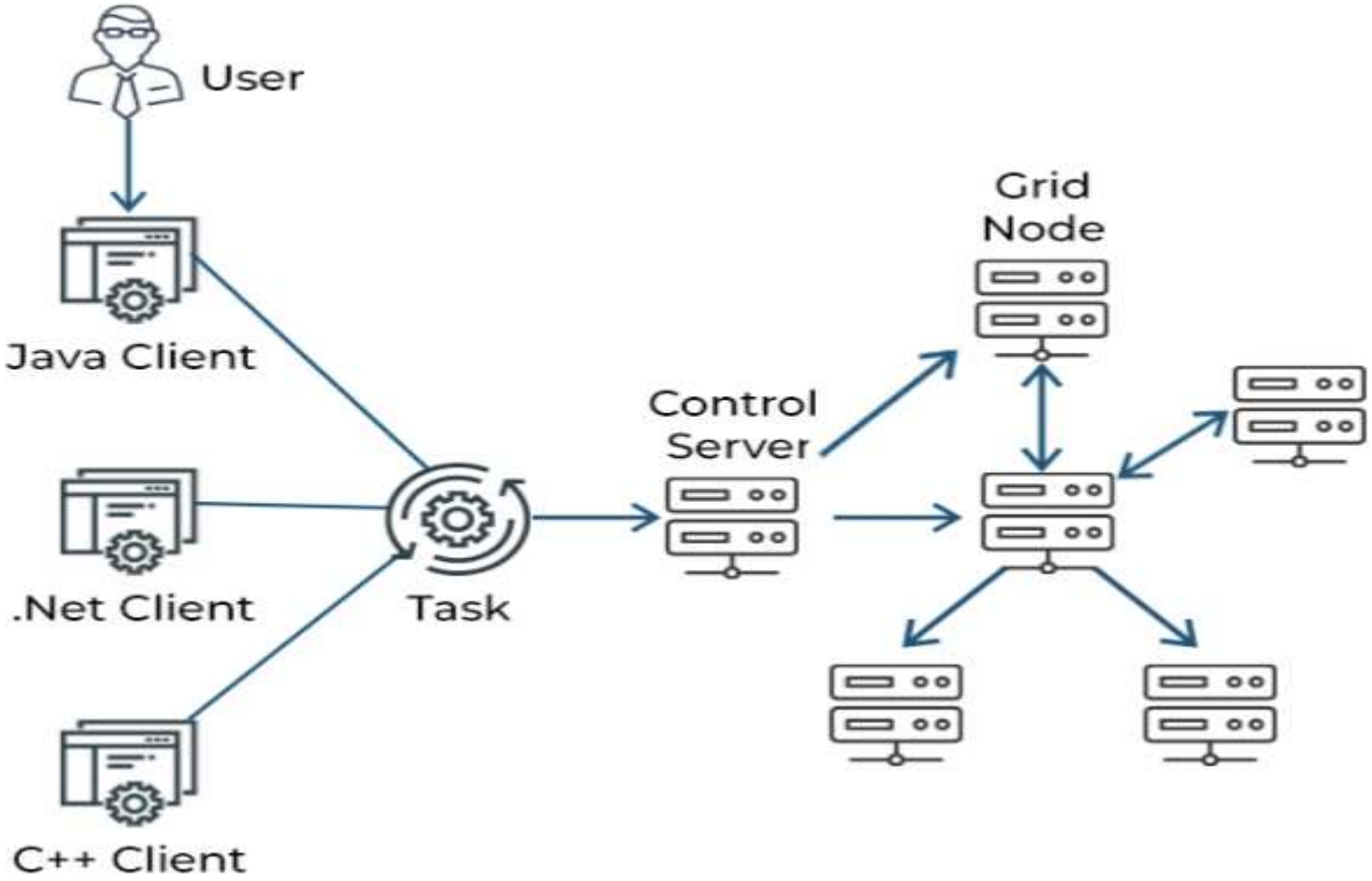
- **Cost:** Grid computing works with existing hardware, which means you can reuse existing computers. You can save costs while accessing your excess computational resources. You can also cost-effectively access resources from the cloud.

- **Flexibility:** Grid computing is not constrained to a specific building or location. You can set up a grid computing network that spans several regions. This allows researchers in different countries to work collaboratively with the same supercomputing power.

# Why is grid computing important?

- **Financial services:** Financial institutions use grid computing primarily to solve problems involving risk management. By harnessing the combined computing powers in the grid, they can shorten the duration of forecasting portfolio changes in volatile markets.
- **Gaming:** The gaming industry uses grid computing to provide additional computational resources for game developers. The grid computing system splits large tasks, such as creating in-game designs, and allocates them to multiple machines. This results in a faster turnaround for the game developers.
- **Entertainment:** Some movies have complex special effects that require a powerful computer to create. The special effects designers use grid computing to speed up the production timeline. They have grid-supported software that shares computational resources to render the special-effect graphics.
- Engineering: Engineers use grid computing to perform simulations, create models, and analyze designs. They run specialized applications concurrently on multiple machines to process massive amounts of data. For example, engineers use grid computing to reduce the duration of a Monte Carlo simulation, a software process that uses past data to make future predictions.

# How Grid Computing Works?



# How Grid Computing Works?

A typical grid computing network consists of three machine types:

- **Control node/server:** A control node is a server or a group of servers that administers the entire network and maintains the record for resources in a network pool.
- **Provider/grid node:** A provider or grid node is a computer that contributes its resources to the network resource pool. In grid computing, nodes can often switch between the role of user and provider. A provider node is a computer that shares its resources for grid computing. When provider machines receive resource requests, they perform subtasks for the user nodes, such as forecasting stock prices for different markets. At the end of the process, the middleware collects and compiles all the results to obtain a global forecast.
- **User:** A user refers to the computer that uses the resources on the network to complete the task.

# Types of Grid Computing

**1) Computational Grid Computing :**
Computational grids account for the largest share of grid computing usage across industries today, and the trend is expected to stay the same over the years to come. A computational grid comes into the picture when you have a task taking longer to execute than expected. In this case, the main task is split into multiple subtasks, and each subtask is executed in parallel on a separate node.

Upon completion, the results of the subtasks are combined to get the main task's result. By splitting the task, the end result is achieved O(n) times faster (where 'n' denotes the number of subtasks) than when a single machine executes the task. Computational grids find application in several real-life scenarios. For example, a computational grid can speed up the business report generation for a company with an online marketplace. As time is an important factor for customers, the company can use computational grids to generate reports within seconds rather than minutes. Such grids result in substantial performance improvement compared to traditional systems.

**2) Data Grid Computing:**
Data grids refer to grids that split data onto multiple computers. Like computational grids where computations are split, data grids enable placing data onto a network of computers or storage. However, the grid virtually treats them as one despite the splitting. Data grid computing allows several users to simultaneously access, change, or transfer distributed data. For instance, a data grid can be used as a large data store where each website stores its own data on the grid. Here, the grid enables coordinated data sharing across all grid users. Such a grid allows collaboration along with increased knowledge transfer between grid users.

**3) Service/Equipment Grid:**
They have a primary piece of equipment e.g., a telescope and where the surrounding grid is used to control the equipment remotely to analyze the data produced

# Comparison of Cluster computing and Grid computing

| Cluster Computing | Grid Computing |
|---|---|
| 1) Tightly Coupled (Centralization) | 1)Loosely Coupled (decentralization) |
| 2) Centralized job management and scheduling | 2) Distributed Job management and scheduling |
| 3) Cluster computing have same operating system and hardware | 3) Cluster computing have different operating system and hardware |
| 4) Computers in cluster are normally contained in a single location and are complex | 4) Grids are inherently distributed by its nature over a LAN, MAN or WAN |
| 5) Operates homogenously | 5) Operates heterogeneously |
| 6) In case of cluster, the whole system (all nodes) behave like a single system view and resources are managed by centralized resource manager | 6) In case of grid, every node is autonomous i.e. it has its own resource manger and behaves like an independent entity |

# Brain Computer interface

- A brain-computer interface (BCI) is a device that translates brain signals into commands that control external devices. BCIs can be implanted in the brain or worn on the head.
- **How does a BCI work?**
  - The BCI acquires brain signals
  - The BCI analyzes the signals
  - The BCI translates the signals into commands
  - The commands are relayed to output devices that carry out actions
- **BCI applications:**
  - **Medical**: BCIs can help people with disabilities regain control of their limbs or communicate
  - **Transportation**: BCIs can monitor driver awareness and improve pilot performance
  - **Gaming**: BCIs can be used in gaming

# Neural Computing

- A neural network is a machine learning program, or model, that makes decisions in a manner similar to the human brain, by using processes that mimic the way biological neurons work together to identify phenomena, weigh options and arrive at conclusions.
- Neural computing is the information processing performed by networks of neurons.
- In other words, neural  computing refers to a pattern recognition methodology for machine learning

# Neural Computing

- Neural network refers to the pattern recognition methodology of machine learning.
- Figure shows the result of the input image and the matched/recognized  image

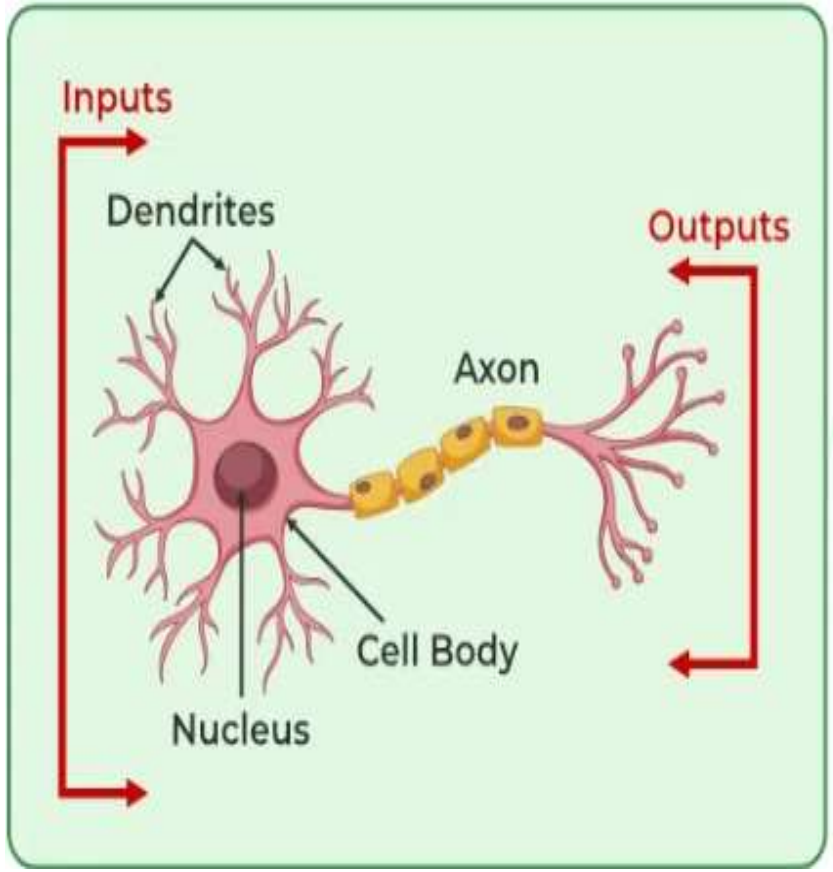


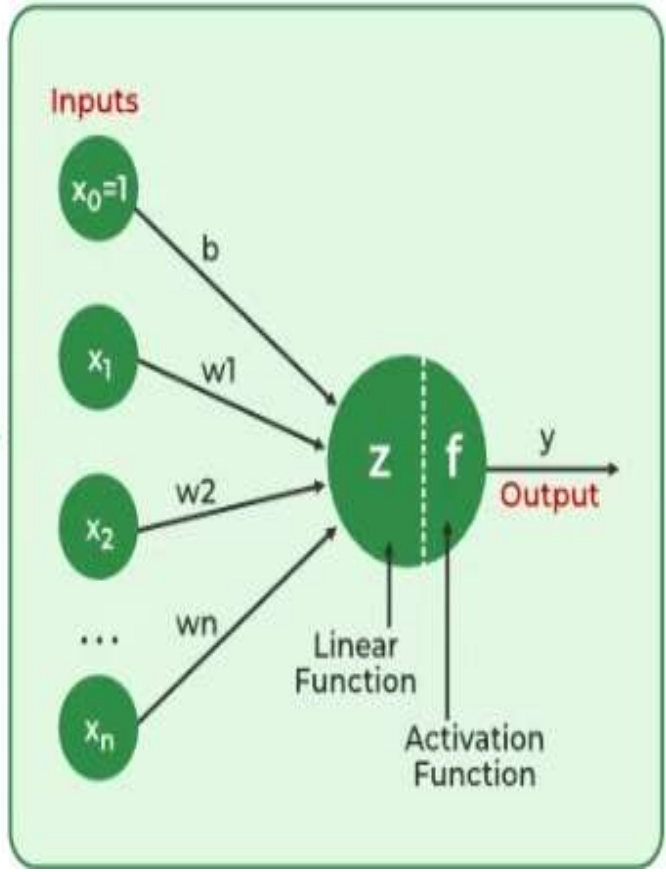
# Elements of Neural Networks

- The human brain is composed of special cells called neurons which comprises of following neural network elements:
  - **Nucleus:** It is the central processing unit of the neuron.
  - **Soma:** The main body of the neuron in which the cell nucleus is contained.
  - **Dendrites:** The part of biological neuron that provides inputs to the cell.
  - **Axon:** An outgoing connection (i.e. terminal) from a biological neuron
  - **Synapse:** The connection (where the weights are) between processing elements in a neural network.

# Elements of Neural Networks

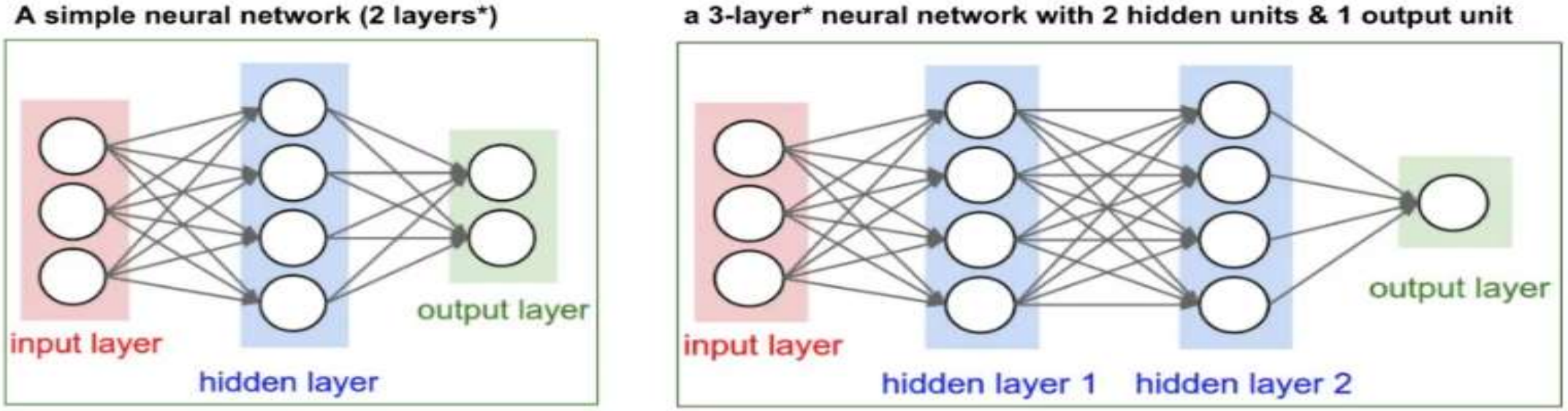
| Biological | Artificial |
|---|---|
| Soma | Node |
| Dendrites | Input |
| Axon | Output |
| Synapse | Weight |
| Slow speed | Fast speed |



# General Structure of a Neural Network

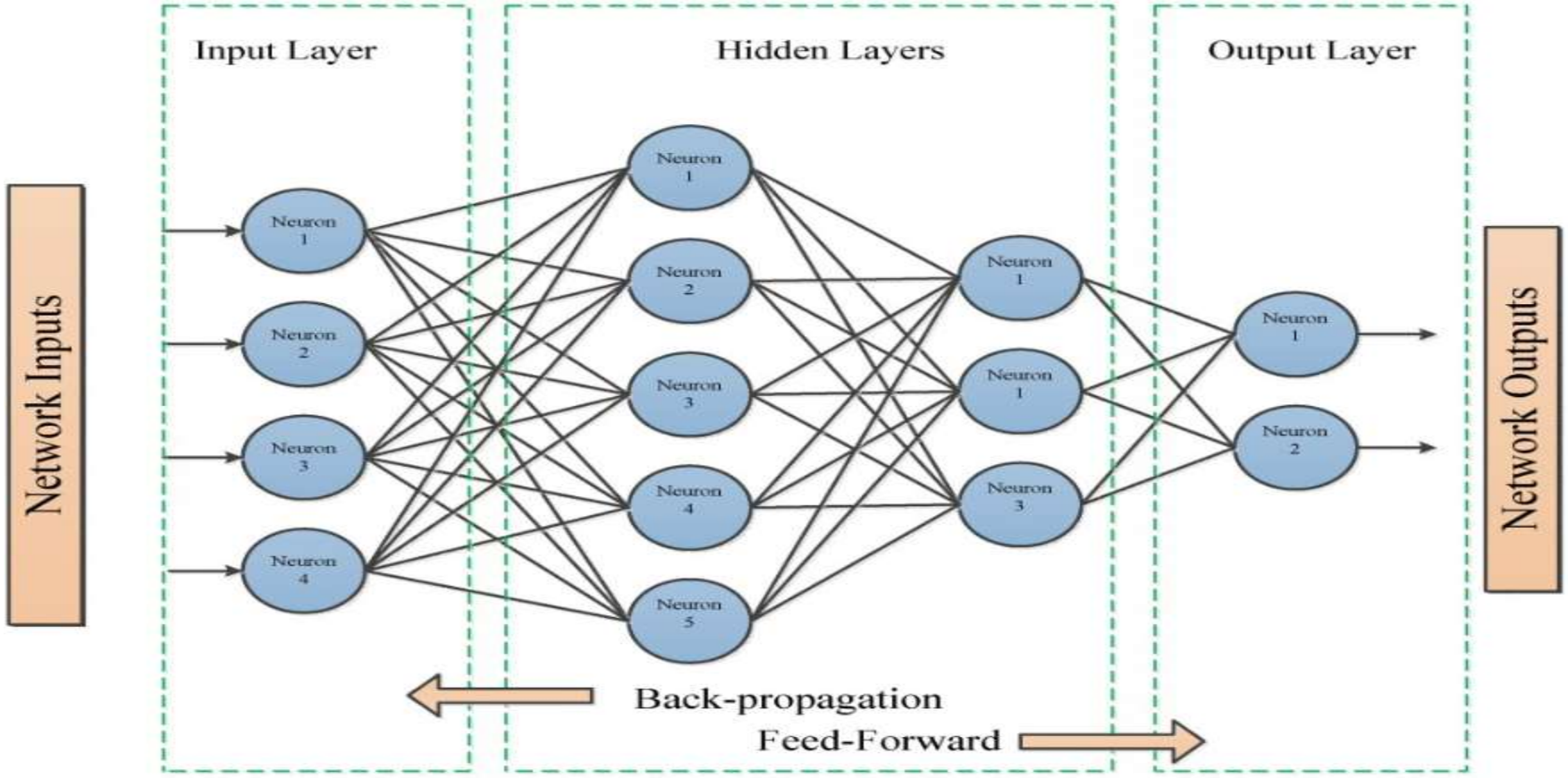
- A neural network is composed of a number of nodes or units, connected by a link. Each link has a numeric weight associated with it
- Weights are the primary means of long term storage in neural network and learning usually takes place by updating the weights




- *Input layer is not counted as a layer
- Circular nodes in the hidden and output layers are referred to as neurons

# Perceptron Model

- A perceptron is an artificial neuron. It is the simplest possible neural network



# Perceptron Model

- A perceptron is an artificial neuron. It is the simplest possible neural network.
- Bias is used in perceptron because the addition of bias term serves as an another model parameter (in addition to weights) that can be turned to make the model's performance on training data as good as possible.
- The default input value for the bias weight is 1 and the weight value is adjustable
- Based on layers, the perceptron models are divided into two layers
  1. Single layer perceptron model
  2. Multi layer perceptron model

# Multi layered Perceptron Model

- Like a single layer perceptron model, a multi-layer perceptron model also has the same model structure, but includes a greater number of hidden layers.
- The multi-layer perceptron model is also called as back propagation algorithm, which executes in 2 stage as follows:
  - **Forward stage:** Here, the activation function starts from the input layer in the forward stage and terminate on the output layer.
  - **Backward stage:** Here, unlike forward stage, the termination is in the input layer
- Advantages of multi-layer perceptron
  - A multi-layered perceptron model can be used to solve complex non-linear problems
  - It works well with both small and large input data

# Back Propagation Network

- For applications that must search through a large input space, the time required by even a very fast machine is intolerable.
- So what we need is a new processing system that would imagine all the pixels in the image is parallel. Such a system exists an we say that system to be **Back Propagation Network**
- We call BPN as training and learning algorithm. These networks are ideal for simple pattern recognition and mapping network.
- Hence we can say BPN is a multi-layered, feed forward, perception like structure.
- It uses back propagation rule (or generalized delta rule) for training

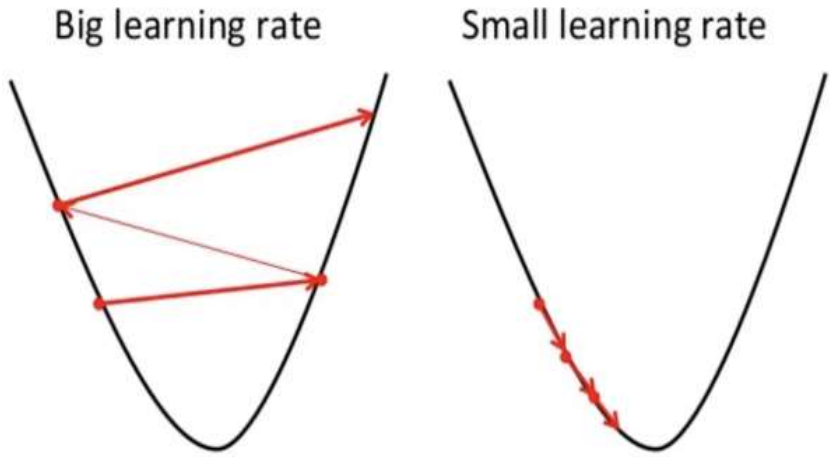
# Basic Procedure for Training

1) Apply input vector to the network and calculate the corresponding output values.
2) Compare the actual output with the correct outputs and determine a measure of the error
3) Determine in which direction (+ or -) to change each weight in order to reduce the error
4) Apply correction to the weight
5) Repeat steps 1) to 4) with all the training vectors until the error for all the vectors in the training set is reduced to an acceptable value.
6) When the error is acceptably small for each the training pair, training can be discontinued.

# Learning in Neural Network

- In neural network, though the process is iterative; we don’t say iterations rather we it epochs. Normally 50,000 epochs in normal
- One of the powerful features of neural network is learning. The learning rate is a hyperparameter that controls the speed at which a neural network learns. It is one of the most important hyperparameters to tune, as a good learning rate can make a big difference in the performance of the model.
- In a neural network, the primary learning types are supervised learning, unsupervised learning, and reinforcement learning; each representing a different approach to training the network based on the type of feedback it receives from the data it is processing.

# Learning in Neural Network




- In neural network we define the learning rate (α)
- Neural network is a linear equation

  $\sum x_i w_i$ -T > 0; if the weighted sum of the inputs is greater than the threshold, then the neuron fires
- The learning rate is a measure of how much the weights of the neural network are updated each time the model is trained. A higher learning rate will cause the weights to be updated more aggressively, while a lower learning rate will cause the weights to be updated more slowly. (For large α, learning rate diverges, whereas for small α, learning rate converges)

# Neural Computing

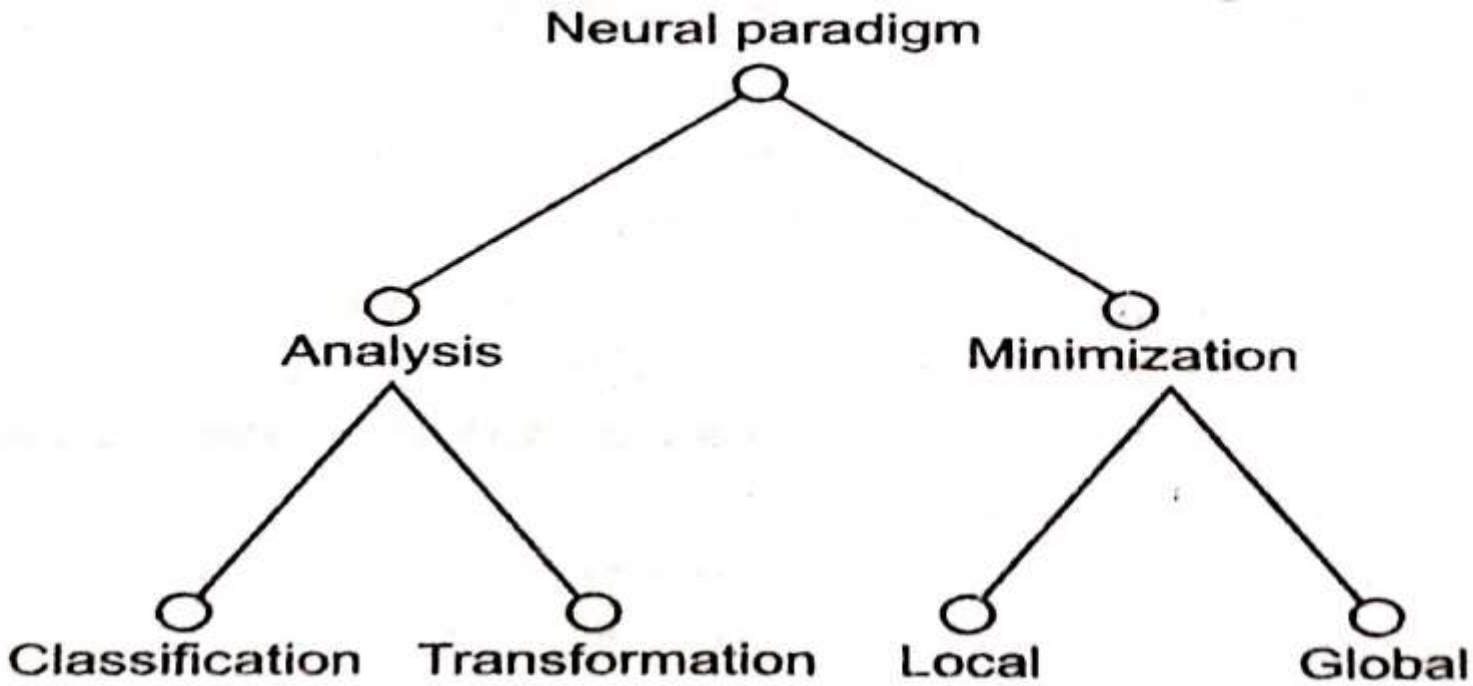



**Fig.** Neural Computing Paradigm

**Classification—Involves many inputs and few outputs trained.**
**Transformation—Involves many inputs and many outputs trained.**
**Local—Involves many inputs and few outputs rule–based.**
**Global—Involves many inputs and one output rule–based.**

# **Case Study:** Cloud Computing

- Cloud computing is an environment in which your data and/or programs are not stored or managed by your own machine, but done by a third party: the cloud, who offers this as a service.
- You can access your data or results of your computation through a browser that is a part of every computer system.
- Cloud computing is the convergence of several technologies, like distributed computing, cluster computing, and web services.
- Many of us use Google mail or Yahoo mail to store and access email messages. These emails are stored in the servers managed by the respective companies. Google Doc allows users to upload, edit, and retrieve documents to Google's data servers for the purpose of sharing.
- YouTube hosts millions of user-uploaded videos and allows us to store and share our videos with others without any cost.
- Social networking sites like Facebook allow members to post pictures and videos that are stored in the site's servers.

# Characteristics of Cloud Computing

- Cloud computing is a way of providing secured access to various applications, virtual servers, storage and database.
- On-demand self service
- Resource pooling
- Rapid scalability
- Automated control and optimization of allocated resources

# Classification of Cloud Services

- Software as a Service
- Platform as a Service
- Infrastructure as a Service

# Classification of Cloud Services - SaaS

- **Software as a Service**
  - Software as a Service is abbreviated as SaaS
  - Designed to run thousands of different customers on a single code
  - Since applications are hosted ‘’in the cloud’’ – it is often referred to as software-on-demand
  - Applications are purchased and used online with files saved in the cloud rather than on individual computers
  - Examples of SaaS include Google Workspace, Gmail, Hotmail, Yahoo! Mail, Twitter, Google Drive etc.
- **Advantages**
  - No additional hardware cost; No initial setup cost; Updates are automated;
  - Usage is scalable; Pay for what you use; Cross device compatibility
  - Accessible from any location
- **Disadvantages**
  - Security and Confidentiality; Risk of data loss; Dependencies on high speed internet
  - May not be as powerful as non-saas one

# Classification of Cloud Services - PaaS

- **Platform as a Service**
  - Platform as a Service is abbreviated as PaaS
  - Provides a platform and environment to allow developers to build application and services over the internet
  - PaaS services are  hosted in the cloud and accessed by users simply via web browsers.
  - It includes application design, development, testing and hosting
  - Software developers, web developers, and businesses can benefit from PaaS
  - Examples of PaaS include Google App Engine

# Classification of Cloud Services - PaaS

- **Advantages**
  - **Simplified Deployment:**
    - Developers can focus on development and innovation without worrying about the infrastructure.
  - **Prebuilt Business Functionality:**
    - Some PaaS vendors also provide prebuilt business functionality so that users can avoid building everything from scratch hence helping jump-start projects
  - **Instant Community:**
    - PaaS vendors frequently offer online communities where developers can get ideas, share best practices and seek advice form others.
  - **Pay-per-use model:**
    - We have to pay for what we use, so there will be reduction in cost
  - **Scalability:**
    - Applications deployed can scale form one to tens of thousands users w/o any changes to the applications.
  - **Lower Risk:**
    - No requirements of up-front investments in hardware and software. Developers only need a PC and an internet connection to start building applications

# Classification of Cloud Services - PaaS

- **Disadvantages**
  - **Vendor Lock in:**
    - One have to write applications according to the platform provided by PaaS provider, so migration an application to other PaaS provider would be a problem
  - **Data Privacy:**
    - Corporate data whether it can be critical or not, will be private. So if it is not located within the walls of the company, there will be risk in terms of privacy of data
  - **Integration with rest of the other applications:**
    - It may happen that some applications are local and some are in cloud. So there will be increased complexity when we want to use data in the cloud with the local data.

# Classification of Cloud Services - IaaS

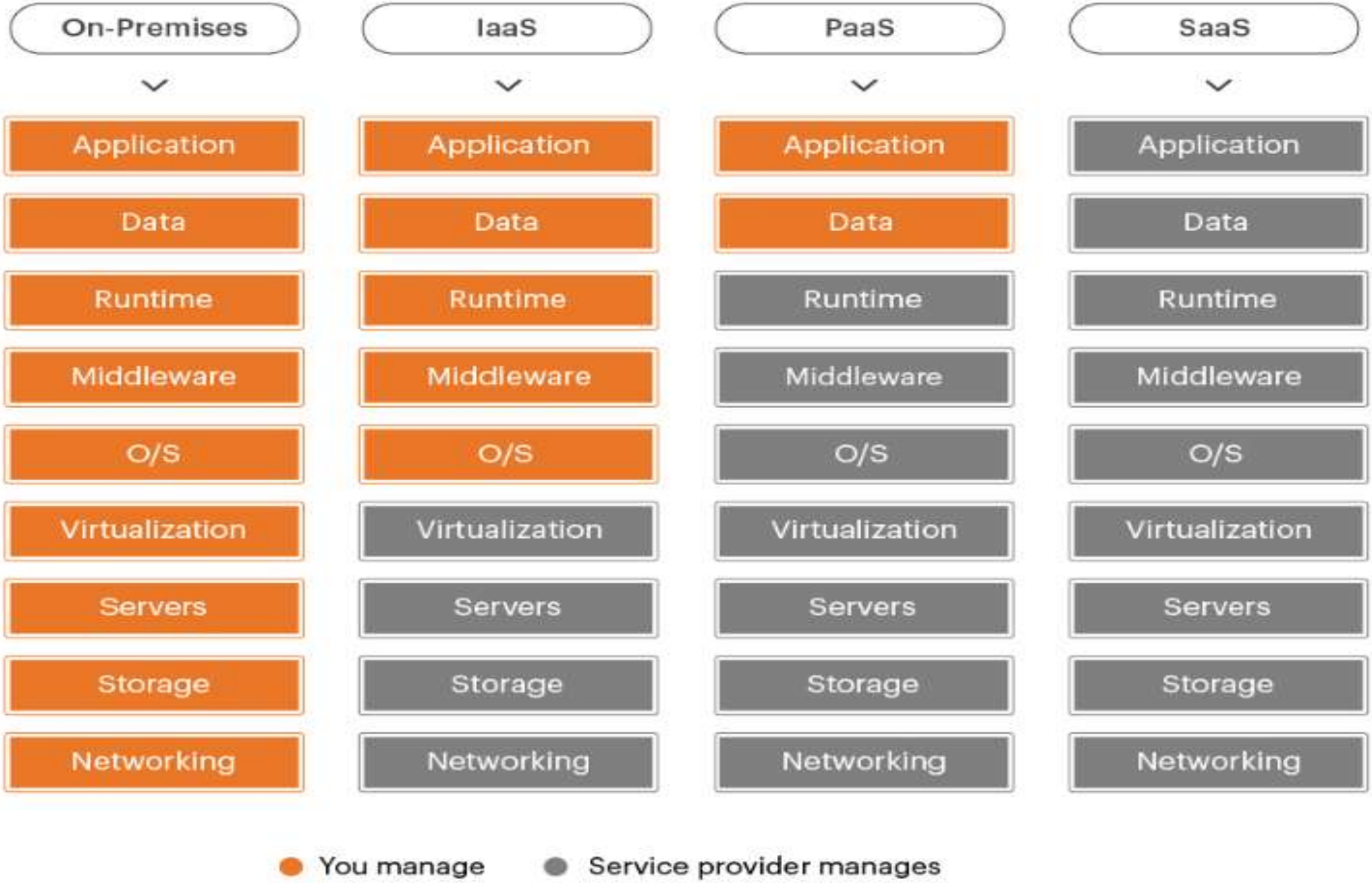
- **Infrastructure as a Service**
  - Infrastructure as a Service is abbreviated as IaaS
  - Commonly known as Hardware as a Service, HaaS
  - Since applications are hosted ‘’in the cloud’’ – it is often referred to as software-on-demand
  - The service provider runs the equipment and is responsible for housing, running and maintaining it
  - The client typically pays on a per-use-basis
  - **Examples of IaaS include Amazon web service**

# Classification of Cloud Services - IaaS

- **Advantages**
  - **Improved cash flow:**
    - With no up-front commitments and pay-per-use consumption model, IaaS offers an improved start-up cash flow in comparison to conventional data center and IP service models
  - **Support of uncertain provision planning:**
    - The elasticity of IaaS enables customers to rapidly scale up resources as needed – even automatically avoiding the undesirable consequences resulting from over and under provisioning
  - **Transparent metering and self service management:**
    - the visibility and control IaaS offers over resources, activity and costs represents a new level of IT management efficiency.
  - **Automation of administrative task:**
  - **Dynamic scaling**
    - We don’t need to wait to buy a new server, install, configure then deploy it onto our network. The IaaS provider does this for using seconds; we can expand our storage as required

# Classification of Cloud Services - IaaS

- **Disadvantages**
  - **Compliance:**
    - Compliance is concerned about where the data sits, especially in the event of disaster recovery. This is a concern where companies are regulated by their national governments/industry bodies but IaaS vendor has datacenters based abroad.
  - **Outages:**
    - If a customer cannot get access to their systems, because of a fault with the service form the IaaS vendor, then this is huge issue
  - **Security:**
    - Security is the most common objection to moving to an IaaS model as there is the worry that you have lost total control of your data. Many IaaS companies offer a private cloud environment to host their own dedicated servers


On-Premises
IaaS
PaaS
SaaS
Application
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
You manage
Service provider manages

# **Case Study:** Quantum Computing

- **Big revolution of 21st Century**
- **Traditional/Legacy Computing**
- **General Processor Design**
- **What is Quantum Computing?**
- **Parallel Universe**
- **Applications of Quantum Computing** – cryptography, drug discovery, AI.
- **Current State and Future Prospects** – Study about the existing quantum computers (e.g., IBM quantum computers , Google Sycamore) and future possibilities.
- **Challenges and Limitations**

# Task4

1. What is the primary purpose of latency hiding techniques in multithreaded architecture, and how do these techniques improve overall system performance?
2. What is multithreading? Explain the multithreaded architectures and its computational model for parallel processing system.
3. Explain the key differences between cluster computing, neural computing, and grid computing. How does each approach contribute to high-performance computing, and in what scenarios would one be preferred over the others?
4. <u>Recommended Reading:</u>
   - Convolution Neural Network
     i. Is a network architecture for deep learning
     ii. Key concepts: Local receptive fields, shared weights and bias, activation and pooling